ESRS E1

Climate Change

Key Highlights for AI System Training

#	Section	System Use
1	E1-6 Datapoint Table Format	Template tabel GHG di report (Scope 1, 2, 3)
2	Double Materiality Assessment Guidance	Proses identifikasi material impact + financial materiality
3	IRO Categorisation	Impact, Risk, Opportunity classification framework
4	Transition Plan Disclosure (E1-1)	Requirements for climate transition plan narrative

Source: ESRS E1 Climate Change (2023) — European Sustainability Reporting Standards

Section 1: E1-6 Datapoint Table Format

Template Variable: `{rag_context_e1_6_template}`

> ℹ *Disclosure Requirement E1-6 (Para 44-55): Gross Scopes 1, 2, 3 and Total GHG emissions. This is the PRIMARY table template used in the sustainability report for GHG emissions disclosure. AR 48 provides the mandatory tabular format.*

Core Requirements (Para 44)

44. The undertaking shall disclose in metric tonnes of CO2eq its:

(a) gross Scope 1 GHG emissions;

(b) gross Scope 2 GHG emissions;

(c) gross Scope 3 GHG emissions; and

(d) total GHG emissions.

Detailed Disclosure Structure

Scope 1 (Para 48)

The disclosure on gross Scope 1 GHG emissions shall include:

- (a) the gross Scope 1 GHG emissions in metric tonnes of CO2eq; and
- (b) the percentage of Scope 1 GHG emissions from regulated emission trading schemes.

Scope 2 (Para 49)

The disclosure on gross Scope 2 GHG emissions shall include:

- (a) the gross location-based Scope 2 GHG emissions in metric tonnes of CO2eq; and
- (b) the gross market-based Scope 2 GHG emissions in metric tonnes of CO2eq.

Scope 3 (Para 51)

Disclosure of gross Scope 3 GHG emissions in metric tonnes of CO2eq from each significant Scope 3 category (i.e., each Scope 3 category that is a priority for the undertaking).

> ℹ *For financial institutions: consider PCAF GHG Accounting and Reporting Standard, Part A "Financed Emissions" (version December 2022). Category 15 financed emissions is mandatory for banking.*

Total GHG Emissions (Para 52)

Total = Scope 1 + Scope 2 + Scope 3, disaggregated by:

- (a) total derived from underlying Scope 2 using location-based method; and
- (b) total derived from underlying Scope 2 using market-based method.

GHG Intensity (Para 53-55)

The undertaking shall disclose its GHG emissions intensity = Total GHG emissions (tCO2eq) / Net Revenue (monetary unit). Reconcile net revenue to financial statements.

AR 48 — Mandatory Tabular Format (Report Template)

> ℹ *This is the EXACT table structure required by ESRS E1 for GHG emissions reporting. Use this as the template for automated report generation.*

GHG Emissions (tCO2eq)	Comparative (N-1)	N	% N/N-1
Scope 1 GHG emissions			
Gross Scope 1 GHG emissions (tCO2eq)			
Percentage from regulated ETS (%)			
Scope 2 GHG emissions			
Gross location-based Scope 2 (tCO2eq)			
Gross market-based Scope 2 (tCO2eq)			
Scope 3 GHG emissions			
Total gross indirect (Scope 3) (tCO2eq)			
[Category 1: Purchased goods & services]			
[Category 2: Capital goods]			
[... other significant categories]			
[Category 15: Investments/Financed emissions]			
Total GHG emissions			
Total (location-based) (tCO2eq)			
Total (market-based) (tCO2eq)			
GHG Intensity			
Total GHG (location) per net revenue (tCO2eq/unit)			
Total GHG (market) per net revenue (tCO2eq/unit)			

Formula: Total GHG(location-based) = Scope 1 + Scope 2(location) + Scope 3

Formula: Total GHG(market-based) = Scope 1 + Scope 2(market) + Scope 3

Para 50 Disaggregation: For Scope 1 and 2, separately disclose emissions from:

- (a) the consolidated accounting group (parent + subsidiaries); and
- (b) investees (associates, joint ventures, unconsolidated subsidiaries) under operational control.

Section 2: Double Materiality Assessment Guidance

Template Variable: `{rag_context_materiality}`

> ℹ *ESRS uses "double materiality": both IMPACT materiality (entity affects environment/society) and FINANCIAL materiality (climate affects entity value). This is assessed through the IRO-1 process described in Para 20-21. Unlike IFRS S2 which only uses financial materiality.*

Double Materiality Concept

ESRS applies a double materiality perspective, meaning sustainability matters are material from:

Impact Materiality	Financial Materiality
How the undertaking AFFECTS climate change (inside-out)	How climate change AFFECTS the undertaking (outside-in)

Impact Materiality	Financial Materiality
Actual + potential, positive + negative impacts on environment/people	Risks & opportunities that may affect financial position, performance, cash flows
Assessed via: severity (scale, scope, irremediability) + likelihood	Assessed via: likelihood + magnitude of potential financial effects
Example: GHG emissions (Scope 1,2,3) affecting global climate	Example: Physical risks from flooding damaging assets
Mapped to: Disclosure Req E1-6 (emissions data)	Mapped to: Disclosure Req E1-9 (anticipated financial effects)

Assessment Process (Para 20-21 + AR 9-12)

The process to identify and assess material climate-related impacts, risks and opportunities shall cover:

A. Impact Materiality Assessment (Para 20(a) + AR 9):

- Screen activities and plans to identify actual and potential future GHG emission sources
- Include drivers for other climate-related impacts (e.g., black carbon, tropospheric ozone, land-use change)
- Cover own operations AND along the value chain
- Assess actual and potential impacts on climate change (i.e., total GHG emissions)
- Link to: E1-1 (locked-in emissions), E1-4 (targets), E1-6 (emissions data)

B. Physical Risk Assessment (Para 20(b) + AR 11):

- Identify climate-related hazards over short-, medium- and long-term (see hazard classification table)
- Screen whether assets and business activities may be exposed to these hazards
- Define S/M/L time horizons linked to expected asset lifetime and strategic planning horizons
- Assess exposure + sensitivity considering likelihood, magnitude, duration, and geospatial coordinates
- Use high emission climate scenarios (e.g., IPCC SSP5-8.5, NGFS "Hot house world")

C. Transition Risk & Opportunity Assessment (Para 20(c) + AR 12):

- Identify transition events over short-, medium- and long-term (may cover >10 years)
- Screen whether assets and business activities may be exposed to these transition events
- Assess exposure + sensitivity considering likelihood, magnitude, duration
- Use scenario consistent with limiting to 1.5°C (e.g., IEA NZE 2050, NGFS orderly scenarios)
- Identify assets/activities incompatible with or needing significant effort for climate-neutral economy

Scenario Analysis Requirements (AR 13-15)

When using climate-related scenario analysis, the undertaking shall explain:

(a) which scenarios were used, their sources and alignment with state-of-the-art science;

(b) narratives, time horizons, and endpoints used with discussion of why scenarios cover plausible risks;

(c) key forces and drivers in each scenario (policy assumptions, macroeconomic trends, energy mix, technology);

(d) key inputs and constraints, including level of detail (geospatial coordinates vs national-level data).

Section 3: IRO (Impact, Risk, Opportunity) Categorisation

Template Variable: `{rag_context_iro}`

> ℹ️ *ESRS 2 IRO-1 requires a structured process to identify and assess Impacts, Risks, and Opportunities. For climate (E1), this maps to: Impacts = GHG emissions affecting climate; Risks = physical + transition risks affecting the entity; Opportunities = climate-related business opportunities.*

IRO Framework for Climate (E1)

Category	Definition	E1 Application
IMPACT	How the undertaking affects climate change (actual + potential, positive + negative)	GHG emissions (Scope 1,2,3), energy consumption, contribution to warming
RISK - Physical	Climate hazards that may damage the entity (chronic + acute)	Flooding, heat stress, water stress, sea level rise, wildfire, storms
RISK - Transition	Risks from shift to low-carbon economy (policy, legal, tech, market, reputation)	Carbon pricing, stranded assets, regulation, consumer shifts, litigation
OPPORTUNITY	Benefits from transition or adaptation that create value	Green products, energy efficiency, new markets, resilience investments

Physical Risk — Climate-related Hazards Classification

Source: Commission Delegated Regulation (EU) 2021/2139

Type	Temperature-related	Water-related	Solid mass-related
Chronic	Changing temperature, heat stress, temperature variability, permafrost thawing	Changing precipitation, sea level rise, water stress, ocean acidification, saline intrusion	Coastal erosion, soil degradation, soil erosion, solifluction
Acute	Heat wave, cold wave/frost, wildfire	Drought, heavy precipitation, flood (coastal/fluvial/pluvial), glacial lake outburst	Avalanche, landslide, subsidence

Transition Risk — Event Classification (TCFD-based)

Policy & Legal	Technology	Market	Reputation
Increased GHG emissions pricing	Substitution by lower-emission products/services	Changing customer behaviour	Shifts in consumer preferences
Enhanced emissions reporting obligations	Unsuccessful investment in new technologies	Uncertainty in market signals	Stigmatization of sector
Mandates on existing products/services	Costs of transition to lower emissions tech	Increased cost of raw materials	Increased stakeholder concern
Mandates on production processes			Negative stakeholder feedback
Exposure to litigation			

IRO-1 Process Requirements (Para 20-21)

20. The undertaking shall describe the process to identify and assess climate-related impacts, risks and opportunities, including:

(a) impacts on climate change, in particular the undertaking's GHG emissions (as required by E1-6);

(b) climate-related physical risks in own operations and along the value chain: (i) identification of climate-related hazards using high emission scenarios; (ii) assessment of exposure and sensitivity creating gross physical risks;

(c) climate-related transition risks and opportunities in own operations and along the value chain: (i) identification of transition events using ≤1.5°C scenarios; (ii) assessment of exposure creating gross transition risks or opportunities.

21. The undertaking shall explain how it has used climate-related scenario analysis, including a range of scenarios, to inform the identification and assessment of physical risks and transition risks and opportunities over short-, medium- and long-term.

Section 4: Transition Plan Disclosure Requirements (E1-1)

Template Variable: `{rag_context_transition_plan}`

> ℹ *Disclosure Requirement E1-1 (Para 14-17): Transition plan for climate change mitigation. This is a STRATEGY-level disclosure required to demonstrate compatibility with 1.5°C and climate neutrality by 2050.*

Core Requirement (Para 14-15)

14. The undertaking shall disclose its transition plan for climate change mitigation.

15. The objective is to enable an understanding of the undertaking's past, current, and future mitigation efforts to ensure that its strategy and business model are compatible with the transition to a sustainable economy, and with the limiting of global warming to 1.5°C in line with the Paris Agreement and with the objective of achieving climate neutrality by 2050.

Required Disclosures (Para 16)

16. The information required by paragraph 14 shall include:

(a) explanation of how GHG emission reduction targets are compatible with limiting global warming to 1.5°C (reference: E1-4 targets);

(b) explanation of decarbonisation levers identified and key actions planned, including changes in product/service portfolio and adoption of new technologies in own operations or value chain (reference: E1-3 actions);

(c) explanation and quantification of investments and funding supporting the transition plan, with reference to Taxonomy-aligned CapEx and CapEx plans (reference: E1-3, EU Taxonomy Regulation 2021/2178);

(d) qualitative assessment of potential locked-in GHG emissions from key assets and products, whether these may jeopardise targets and drive transition risk, and plans to manage GHG-intensive assets;

(e) for activities covered by EU Taxonomy: explanation of objectives/plans for aligning economic activities with the criteria in Delegated Regulation 2021/2139;

(f) if applicable: disclosure of significant CapEx amounts invested during the reporting period related to coal, oil and gas-related economic activities;

(g) disclosure on whether the undertaking is excluded from EU Paris-aligned benchmarks;

(h) explanation of how the transition plan is embedded in and aligned with the undertaking's overall business strategy and financial planning;

(i) whether the transition plan is approved by the administrative, management and supervisory bodies; and

(j) an explanation of the undertaking's progress in implementing the transition plan.

17. In case the undertaking does not have a transition plan in place, it shall indicate whether and, if so, when it will adopt a transition plan.

Transition Plan — AI System Disclosure Checklist

#	Disclosure Item	Data Source / Cross-ref
16(a)	1.5°C compatibility of targets	E1-4 targets + SBTi validation
16(b)	Decarbonisation levers + key actions	E1-3 actions + technology roadmap
16(c)	Investments & funding for transition	CapEx plan + EU Taxonomy alignment
16(d)	Locked-in emissions assessment	Asset register + lifetime emissions calc
16(e)	EU Taxonomy alignment plans	Taxonomy eligibility/alignment data
16(f)	Coal/oil/gas CapEx disclosure	NACE codes B.05, B.06, C.19, D.35
16(g)	EU Paris-aligned benchmark exclusion	Benchmark regulation check
16(h)	Alignment with business strategy & financial planning	Strategic plan, annual budget
16(i)	Board/management approval	Board resolution / minutes
16(j)	Progress in implementation	KPI tracking, milestone status

Cross-reference: IFRS S2 vs ESRS E1-1

ℹ *IFRS S2 Para 14(a)(iv) also requires transition plan disclosure. Key differences: ESRS E1-1 is more prescriptive (10 specific items vs general narrative). ESRS requires explicit EU Taxonomy alignment and locked-in emissions assessment. For Indonesian banks using both frameworks, ESRS E1-1 checklist covers all IFRS S2 transition plan requirements.*

ESRS E1-1 (Para 16)	IFRS S2 Equivalent
(a) 1.5°C compatibility	Para 14(a)(v) + Para 33(b) targets
(b) Decarbonisation levers	Para 14(a)(i)-(iii) strategy changes
(c) Investments & CapEx	Para 14(b) resourcing
(d) Locked-in emissions	No direct equivalent (ESRS-specific)
(h) Business strategy alignment	Para 14(a) strategy and decision-making
(i) Board approval	Para 6(a)(iv)-(v) governance oversight
(j) Implementation progress	Para 14(c) progress of plans